

Descension of Additive Functors to Chain Categories

Setting:

- We have an additive functor $F: A \rightarrow B$ between abelian categories

Statement:

Additive functors preserve the chain complex condition, and further preserve chain homotopies between chain maps, thus giving a ladder of induced functors as shown at bottom right.

Proof:

This result is demonstrated in two steps:

- ① additive functors preserve chain complexes
- ② additive functors preserve chain homotopies

$$\begin{array}{ccc} A & \xrightarrow{F} & B \\ \downarrow & & \downarrow \\ \text{Ch } A & \xrightarrow{\bar{F}} & \text{Ch } B \\ \downarrow & & \downarrow \\ H\text{Ch } A & \xrightarrow{\check{F}} & H\text{Ch } B \end{array}$$

①: Take a chain complex $C_\bullet$ in $\text{Ch}A$. The chain condition states $d_n d_{n+1} = \emptyset$ for all n , and acting F on this equality yields:

$$\begin{array}{ccccccc} \cdots & \rightarrow & C_{n+1} & \xrightarrow{d_{n+1}} & C_n & \xrightarrow{d_n} & C_{n-1} \xrightarrow{d_{n-1}} \cdots \\ & & F \downarrow & & F \downarrow & & F \downarrow \\ \cdots & \rightarrow & FC_{n+1} & \xrightarrow{Fd_{n+1}} & FC_n & \xrightarrow{Fd_n} & FC_{n-1} \xrightarrow{Fd_{n-1}} \cdots \end{array}$$

$$F(d_{n+1} d_n) = F(\emptyset)$$

The LHS will clearly simplify via either covariant or contravariant functoriality, but we need $F(\emptyset)$ to be $\emptyset$ in order to result in a valid chain complex in $\text{Ch}B$.

Since we are working in an abelian category, our hom-sets are in fact hom-groups, abelian ones. Additivity means the morphism assignment of F is an abelian group homomorphism, thus we get $F(\emptyset + f) = F(\emptyset) + F(f) = F(f) + F(\emptyset)$. Thus $F(\emptyset)$ satisfies the conditions for the additive identity, so uniqueness gives $F(\emptyset) = \emptyset$. Whether F is co/contra-variant matters not, for we yield a complex either way, just reversed in the contravariant case.

For F to validly descend to a chain functor, chain maps must be sent to chain maps, which means we need square-commutativity. Consider the arbitrary commuting square in A pictured centrally below.

$$\begin{array}{ccc}
 \begin{array}{ccc}
 FB' & \xrightarrow{F\beta} & FB \\
 Ff' \downarrow & & \downarrow Ff \\
 FA' & \xrightarrow{F\alpha} & FA
 \end{array} & \xleftarrow{F} & \begin{array}{ccc}
 A & \xrightarrow{f} & B \\
 \alpha \downarrow & G & \downarrow \beta \\
 A' & \xrightarrow{f'} & B'
 \end{array} \\
 \text{contravariant} & & \text{covariant}
 \end{array}
 \quad \Rightarrow \quad
 \begin{array}{ccc}
 FA & \xrightarrow{Ff} & FB \\
 F\alpha \downarrow & & \downarrow F\beta \\
 FA' & \xrightarrow{Ff'} & FB'
 \end{array}$$

Symbolically, this square's commutativity means $\beta \circ f = f' \circ \alpha$, and thus acting F on this yields the two cases:

contravariant

$$\begin{aligned}
 F(\beta \circ f) &= F(f' \circ \alpha) \\
 \Downarrow \\
 Ff \circ F\beta &= F\alpha \circ Ff'
 \end{aligned}$$

covariant

$$\begin{aligned}
 F(\beta \circ f) &= F(f' \circ \alpha) \\
 \Downarrow \\
 F\beta \circ Ff &= Ff' \circ F\alpha
 \end{aligned}$$

Thus we get commutativity of the appropriate squares for either variance. This gives us valid descension of $F: A \rightarrow B$ to $\bar{F}: \text{Ch}A \rightarrow \text{Ch}B$.



②: Consider a chain homotopy s between chain maps $f, g: C_\bullet \rightarrow C'_\bullet$ in $\text{Ch } A$. This means that for all n ,

$$f_n - g_n = d'_{n+1} s_n + s_{n-1} d_n.$$

$$\begin{array}{ccccccc} \cdots & \rightarrow & C_{n+1} & \xrightarrow{d_{n+1}} & C_n & \xrightarrow{d_n} & C_{n-1} \xrightarrow{d_{n-1}} \cdots \\ & & \searrow s_n & & \downarrow h_n & & \swarrow s_{n-1} \\ \cdots & \rightarrow & C'_{n+1} & \xrightarrow{d'_{n+1}} & C'_n & \xrightarrow{d'_n} & C'_{n-1} \xrightarrow{d'_{n-1}} \cdots \end{array}$$

Applying F to this equality and employing additivity yields:

$$F(f_n - g_n) = F(d'_{n+1} s_n + s_{n-1} d_n) \Rightarrow Ff_n - Fg_n = F(d'_{n+1} s_n) + F(s_{n-1} d_n)$$

In the covariant case we are done since we just take Fs_n as our prospective homotopy; both $\bar{F}f$ and $\bar{F}g$ are valid chain maps and Fd_n is the differential of the resulting complex, so we immediately acquire the desired relation.

In the contravariant case the equality changes slightly, but our prospective homotopy is still just Fs_n . Below we see that the homotopy condition gets the addition reversed, but commutativity ensures it's still valid. Thus additive functors of either variance preserve homotopies, and so induce functors on homotopic chain complex categories.

$$\begin{array}{ccccccc} \cdots & \rightarrow & FC'_{n-1} & \xrightarrow{Fd'_{n-1}} & FC'_n & \xrightarrow{Fd'_n} & FC'_{n+1} \rightarrow \cdots \\ & & \swarrow Ff_{n-1} & & \downarrow Ff_n & & \searrow Fg_n \\ \cdots & \rightarrow & FC_{n-1} & \xrightarrow{Fd_{n-1}} & FC_n & \xrightarrow{Fd_n} & FC_{n+1} \rightarrow \cdots \end{array}$$

$$Ff_n - Fg_n = F(d'_{n+1} s_n) + F(s_{n-1} d_n) = Ff_n \circ Fd'_{n+1} + Fd_n \circ Ff_{n-1}$$